T14

$\xi \sim N(a, \sigma_x^2)$ $\quad \sigma_x^2 = 2$ $\quad n = 3$ $\quad \alpha = 0{,}05$

$\eta \sim N(b, \sigma_y^2)$ $\quad \sigma_y^2 = 1$ $\quad m = 2$

$\vec{x}_n = \{-1{,}11;\ -6{,}10;\ 2{,}42\}$ $\quad \vec{y}_m = \{-2{,}29;\ -2{,}91\}$

$H_0: a = b$

$H_1: a > b$

по Т Фишера ① $\dfrac{\bar{x} - a}{\sigma_x}\sqrt{n} \sim N(0,1)$

② $\dfrac{\bar{y} - b}{\sigma_y}\sqrt{m} \sim N(0,1)$

① $\bar{x} - a \sim N\left(0, \dfrac{\sigma_x^2}{n}\right)$ ② $\bar{y} - b \sim N\left(0, \dfrac{\sigma_y^2}{m}\right)$

$\bar{x} \sim N\left(a, \dfrac{\sigma_x^2}{n}\right)$ $\quad \bar{y} \sim N\left(b, \dfrac{\sigma_y^2}{m}\right)$

Выборки независимые $\Rightarrow$

$$\Rightarrow \quad \bar{x} - \bar{y} \sim N\left(a-b, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)$$

$$\bar{x} - \bar{y} \sim N\left(a-b, \frac{7}{6}\right)$$

или $$\frac{\bar{x} - \bar{y} - (a-b)}{\sqrt{7/6}} \sim N(0,1)$$

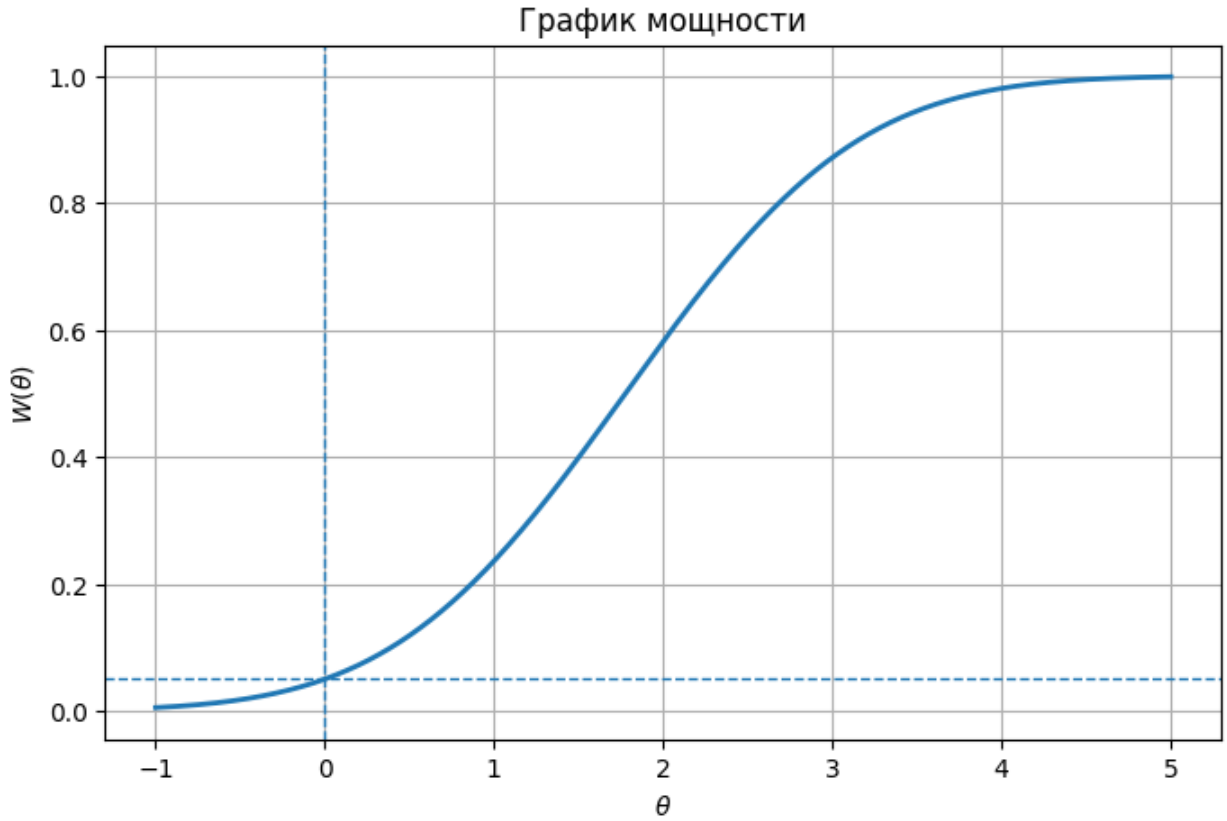
$$\frac{\bar{x} - \bar{y} - (a-b)}{\sqrt{7/6}} \geqslant u_{1-\alpha}$$

$H_0: a = b$ $\Rightarrow$ она отвергается, если

$$G: \frac{\bar{x} - \bar{y}}{\sqrt{7/6}} \geqslant u_{1-\alpha}$$

Обозначим $\theta = a - b$

$$W(\theta) = P\left(\frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \in G \,\middle|\, H_1\right) = 1 - \Phi\left(u_{1-\alpha} - \frac{\theta}{\sqrt{7/6}}\right)$$



$\bar{x} - \bar{y} \approx 1{,}0033$

$u_{1-\alpha} = u_{0{,}95} \approx 1{,}645$

$\dfrac{\bar{x} - \bar{y}}{\sqrt{7/6}} \approx 0{,}93$

$0{,}93 < 1{,}645$

$\Rightarrow \notin G \rightarrow$

нет оснований отвергать H_0